MODULE 1

1. Explain the voltage divider bias circuit analysis with a neat diagram.
2. Explain the emitter follower bias with a neat circuit.
3. Explain the base biased amplifier with waveforms.
4. Explain the Voltage divider bias amplifier with waveforms.
5. Draw the Ebers moll model and π model for NPN transistor and explain.
6. Draw the ac and dc equivalent circuit for using π model and derive the gain equation.
7. Explain the common collector amplifier with waveforms.
8. Draw the ac and dc equivalent circuit for using π model for TSE and explain.
9. Derive the input and output impedance equation for emitter follower circuit.

MODULE 2

1. Explain the fixing Vg biasing method.
2. Discuss the small signal operation of MOSFET.
3. List the characteristics parameters of amplifiers.
4. Draw the T-equivalent circuit model for MOSFET
5. Explain the common source amplifier without source resistor.
6. Discuss the common source amplifier with source resistor with a neat circuit diagram.
7. Explain the common gate amplifier with help of circuit diagram.
8. Explain the source follower amplifier.

MODULE 3

1. Explain the summing amplifier with a neat diagram.
2. Explain the summing amplifier as a sub-tractor and average.
3. Explain the binary weighted D/A converter with a neat diagram
4. Explain the R-2R ladder D/A converter with a neat circuit
5. Discuss the comparator with zero and without zero reference with a hysteresis.
6. With a neat diagram explain the wein bridge oscillator.
7. Explain the RC phase shift oscillator.
8. Explain the Zero crossing detector.
9. Explain the single supply comparator and open collector comparator.
10. Explain the Schmitt trigger with Hysteresis.
11. Describe the colpitts oscillator.
12. Explain the Hartley oscillator and crystal oscillator with a circuit diagram.
13. Describe the 555 timer with its functional diagram.

14. Discuss the monostable operation of 555 timer.
15. Explain the astabe operation of 555 timer.
16. Explain the voltage controlled oscillator with a neat diagram.

MODULE 5

1. Explain the classification of power amplifiers.
2. Discuss the two different types of load lines of VDB amplifier with diagrams.
3. Explain the class A power amplifier with load lines. .
4. Discuss the class B amplifier operation.
5. Describe the class B push pull emitter follower amplifier with load line.
6. Discuss the class C operation.
7. Differentiate b/w class A, Class B, Class C amplifier.
8. Discuss the four layer diode/Shockley diode with neat diagram.
9. Describe the different turn-on methods of SCR.
10. Explain the SCR phase control operation.
11. Explain the different turn-Off methods of SCR.
12. Explain the diac and triac thyristors.
13. Explain the triac phase control operation.
14. Explain the construction of IGBT with its operation.
15. Explain the UJT, photo SCR, PUT, Gate controlled switch, silicon controlled switch.